

Problem Statement and Goals

Squigglr

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Table 1: Revision History

Date	Developer(s)	Change
26 September 2026	Rayan Nasrallah	Added the Problem Statement section, covering the problem, inputs and outputs, stakeholders, and environment, with supporting references.
26 September 2026	Rayan Nasrallah	Added the reflection responses to the appendix and the generative AI use disclosure.

1 Problem Statement

1.1 Problem

Visual-motor integration is the ability to coordinate visual information with movement. Children rely on it when drawing, copying shapes, and learning to write. Research links these skills to handwriting legibility and identifies motor or handwriting difficulties in some children with attention-deficit/hyperactivity disorder (ADHD) or autism [1–3]. The need extends from kindergarten and primary school learners to older children who require continued support.

Children need opportunities to practise at an appropriate level, understand their difficulties, and develop greater independence. Parents and educators need to distinguish what a child can do independently from what they accomplish with assistance. A finished drawing alone does not provide that distinction.

Research on structured foundational activities supports the value of practice for visual-motor development [4].

There is an opportunity to build a system that supports drawing practice suited to individual abilities, provides understandable feedback, and helps children and supporting adults track independent progress. The intended outcome is improved drawing skills and greater independence, with opportunities for creative expression. Supporting adults need access to both individual attempt records and understandable summaries of change over time. Clinical diagnosis and treatment are outside the project scope.

1.2 Inputs and Outputs

1.2.1 Inputs

- Selected drawing activities and support preferences.
- Children’s drawing attempts during lessons and creative activities.
- Editing actions, such as moving, resizing, recolouring, or erasing drawing elements.

1.2.2 Outputs

- Drawing targets, demonstrations, and guidance appropriate to the activity and support level.
- Simple activity feedback and completion results based on predefined observations.
- Editable creative artwork, including supported automatic refinements and recoverable original drawings.
- Attempt records showing encountered difficulties, changes within an attempt, guidance used, and the final outcome.
- Progress summaries showing changes across attempts and recurring difficulties, with access to the supporting records.

1.3 Stakeholders

- Kindergarten and primary school children.
- Children of different ages who need additional support, including children with autism, ADHD, or other developmental or learning differences.
- Parents, guardians, and caregivers.
- Educators and learning-support staff.
- School administrators.

- Domain advisers in inclusive education and accessible educational software.
- Dr. Christopher Anand, project supervisor.

1.4 Environment

Users will need access to a supported web browser on a compatible device with a display and a supported drawing input. Using the application will not require installing a separate application.

2 Goals

3 Stretch Goals

[Goals you hope to get to, but not necessary. They are also helpful if the scope of the original project needs to be expanded. —SS]

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

Rayan: Through my parents, I connected with a specialist who holds a doctorate and develops software for children with autism, as well as the head of a nonprofit organisation serving people with additional support needs, including autism and ADHD. These connections gave us a clearer direction for identifying stakeholders who could inform our requirements. They also helped me recognise the value of seeking both software-design expertise and experience supporting our intended users. We now have potential contacts who can help us question our assumptions and understand needs that we might otherwise overlook.

2. What pain points did you experience during this deliverable, and how did you resolve them?

Rayan: When I first presented our idea to Dr. Smith, he explained that we needed to justify the need for the application. I realised that describing useful features was not enough; we needed evidence supporting the problem we wanted to address. I reviewed research on visual-motor development, drawing practice, and difficulties experienced by some children with autism or ADHD. Comparing these findings helped me connect our proposal to existing work and explain the rationale for structured drawing practice. It also clarified the distinction between evidence supporting our approach and evidence that our particular application will be effective. In future deliverables, I would establish the evidence and its limitations earlier so that they guide our scope from the beginning.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Team response: We narrowed the core scope to structured drawing practice, progression towards independent drawing, adjustable support, free drawing with refinement, attempt recording, and parent-facing progress

summaries. We moved camera-based tracing and letter activities to stretch goals so that they would not distract from completing the core experience. We also limited activity feedback to predefined observations, such as missed tracing points, rather than personalised generated advice.

We distinguished user goals from supporting requirements such as account creation and browser compatibility. We also separated recording individual attempts from summarising progress, giving each goal a distinct purpose. Our completion checks now use prepared drawing attempts and activity histories that the team can evaluate without depending on child participants. These decisions keep the project manageable while retaining substantial design work in drawing assessment, adaptive guidance, editable artwork, and accurate progress reporting.

Generative AI Use Disclosure

Our team used OpenAI GPT-6 Astra to improve the presentation of this document. We independently developed the project ideas and wrote the initial content before using AI. Its role was limited to L^AT_EX formatting, citation formatting, grammar correction, and improvements to sentence structure, clarity, and flow. No additional tools or skills were used.

All project ideas and substantive decisions originated with the team. We reviewed the suggested edits against our original draft to ensure that they preserved our intended meaning and checked the formatted citations against the source materials.

References

1. Daly, C. J., Kelley, G. T., and Krauss, A. (2003). Relationship between visual-motor integration and handwriting skills of children in kindergarten: A modified replication study. *American Journal of Occupational Therapy*, 57(4), 459–462. <https://doi.org/10.5014/ajot.57.4.459>
2. Kaiser, M.-L., Schoemaker, M. M., Albaret, J.-M., and Geuze, R. H. (2015). What is the evidence of impaired motor skills and motor control among children with attention deficit hyperactivity disorder (ADHD)? Systematic review of the literature. *Research in Developmental Disabilities*, 36, 338–357. <https://doi.org/10.1016/j.ridd.2014.09.023>
3. Fuentes, C. T., Mostofsky, S. H., and Bastian, A. J. (2009). Children with autism show specific handwriting impairments. *Neurology*, 73(19), 1532–1537. <https://doi.org/10.1212/WNL.0b013e3181c0d48c>
4. Taverna, L., Tremolada, M., Dozza, L., Zanin Scaratti, R., Domahs, U., Lallo, C., and Toso, B. (2020). Who benefits from an intervention program on foundational skills for handwriting addressed to kindergarten children and first graders? *International Journal of Environmental Research and Public Health*, 17(6), 2166. <https://doi.org/10.3390/ijerph17062166>